AI PRODUCT CASE STUDY · PAYMENTS · INDIA + US

Pause: stopping scam payments without slowing down honest ones

How I would build a real-time scam-risk and intervention layer for UPI in India and for Zelle, RTP and FedNow in the US

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In an authorized push payment scam, the victim does everything right by the bank's rules. They unlock their own phone, type their own PIN and send the money themselves. Every security check passes, and the money is gone in seconds.

In April 2026 the RBI asked whether every person-to-person payment above ₹10,000 should be held for an hour.[1] This case study is my answer: hold the few payments that look like a scam in progress, explain why in plain words, and let everything else go through instantly.

At a glance	
The problem	India's cybercrime portal logged 28 lakh digital payment fraud reports worth ₹22,931 crore in 2025.[1] Americans reported about $16 billion in fraud losses to the FTC in 2025.[13]
The bet	Scams leave context behind: an active call, a new payee, a risky phone number, an odd amount. A model can read that context in under 100 ms. A blanket rule can't.
The product	A risk score plus a five-step intervention ladder, from "do nothing" to "decline", with warnings that name the specific scam.
North star	Scam value stopped per month, kept honest by a good-user interrupt rate and P2P completion.
MVP	One partner app, Android, P2P payments to new payees, Hindi, Telugu and English, shadow mode first.

1 Why I picked this problem

I've spent the last two and a half years building AI for motor insurance at ICICI Lombard. From my work on six ML fraud risk models in our claims workflows, and on voice analytics that caught inconsistencies in customer and surveyor calls, I learned two things that carry straight over. Fraud is rarely one red flag; it's a few weak signals that only look suspicious together. And every flag costs an honest customer something: in claims a slower settlement, in payments a rent transfer that doesn't go through.

APP scams are the hardest version of this, because the customer really is the one paying. There's no stolen password to catch. So the question isn't only "can we spot it?" It's "what do we show a frightened person, at the moment of payment, that actually makes them stop?"

2 The problem, in numbers

	India	United States
Scale	UPI: 24.07 bn transactions worth ₹29.37 lakh crore in Sep 2026, about 802 mn a day[2]	Zelle: 4.2 bn transactions, $1.2 tn in 2025.[11] RTP: 128 mn transactions, $480 bn a quarter[17]
Reported fraud	28 lakh digital payment fraud reports, ₹22,931 crore, in 2025 (24 lakh, ₹22,848 crore in 2024)[1]	About $16 bn reported to the FTC in 2025, $3.5 bn of it to imposter scams[13]
Signature scam	"Digital arrest": 1,23,672 cases and ₹1,935 crore lost in 2024, up from 39,925 cases in 2022[4]	Bank and government imposters. Over-60s reported $7.7 bn of losses to the FBI in 2025, up 37%[14]
Where the money is	Payments above ₹10,000: 45% of reported cases, 98.5% of the value[1]	0.02% of Zelle payments draw a fraud or scam report,[12] roughly 840,000 a year (my arithmetic)
Regulators	RBI paper on APP fraud (Apr 2026); compensation for small frauds, coercion included, from Jan 2027[1, 7]	CFPB sued Zelle's operator and three banks, then dropped it (2025).[15] New York's AG case is going ahead[16]

The loss is concentrated. Almost all the money sits in a minority of larger payments, so friction can go where the value is.

The numbers don't agree. The Finance Ministry told Parliament that UPI fraud in FY2024–25 was 12.64 lakh cases worth ₹981 crore.[3] The cybercrime portal counted ₹22,848 crore of digital payment fraud in 2024.[1] Different scopes explain part of that 20x gap. For an AI PM it matters more than any market size: the model learns whichever definition you feed it, so labels are a product decision (section 7).

The UK is a useful preview. Since October 2024 payment firms there must reimburse APP scam victims. A year in, 88% of money claimed back was returned and claims were about 15% lower, which the regulator reads as firms stopping more scams up front.[18] Liability moved, and prevention followed.

3 Market sizing

I size this as value at stake, not as a software market: how much of the reported loss in section 2 a payment-moment product can reach.

Layer	India	United States
Reachable at the moment of payment	Payments above ₹10,000: ~98.5% of reported value[1]	~$2 bn of imposter losses sent by bank transfer or payment (unverified)[13]
Buyers	756 banks live on UPI[2]; three apps carry 90%+ of volume[5]	~2,400 banks and credit unions on Zelle[12]; 1,725 on FedNow[17]

Year-1 obtainable (my estimate). In India, one partner with ~5% of UPI volume that stops 10% of its scam value keeps about ₹115 crore a year in customers' accounts (₹22,931 crore × 5% × 10%). That's an order-of-magnitude check, not a forecast. In the US, I'd size one mid-size bank or credit union group from shadow-mode data on its own traffic.

4 Who gets scammed, and how

Persona (composite)	What happens	What would actually help
Ramesh, 68 **Retired, Hyderabad**	A "CBI officer" on video says his Aadhaar is tied to money laundering. Don't hang up, don't tell family, move your savings to a "verification account". Three payments in two days.	Something that breaks the isolation, and permission to stop.
Sneha, 27 **Engineer, Bengaluru**	A Telegram "task job" pays ₹500, then asks ₹5,000, ₹20,000, ₹60,000 to unlock bigger tasks. Each payment goes to a new UPI ID.	Someone who names the pattern before the third payment.
Linda, 74 **Retired teacher, Ohio**	Her "bank's fraud team" calls from a spoofed number and tells her to Zelle money to a "safe account in her name".	A way to check in her app whether her bank is really on the line.
Priya **Bank fraud analyst**	Works the queue of held payments and calls customers back.	Reason codes and a scam type, not a bare score. One-tap outcomes.

5 Why AI here, and where not

Rules already exist, and they're blunt. NPCI switched off P2P collect requests in October 2025 to kill one scam pattern.[6] The RBI's proposed lag would delay every P2P payment above ₹10,000 by an hour,[1] rent and hospital deposits included. The RBI itself asks, in Question 4, whether the lag should instead depend on the risk the payer's bank sees in each transaction.[1] That question is the opening for this product.

Where AI earns its place

- **Combining weak signals.** An active call, a new payee, an amount six times the user's normal, a payee number the telecom department has flagged. Each is common alone. Together they're a scam in progress. A model weighs dozens of these better than a rule list.
- **Naming the scam.** The model predicts the likely type (impersonation, task or investment, marketplace, romance) so the warning can be specific. Google Pay already shows over a million fraud warnings a week in India,[10] so the bar is warnings people act on, not more of them.
- **Writing warnings at scale.** An LLM drafts and translates copy offline, people review it, and the app serves approved templates. No live generated text on the payment screen.

Where I wouldn't use AI

- **Hard stops.** A payee number rated "Very High" risk by the Department of Telecom's Financial Fraud Risk Indicator is declined by rule, as PhonePe already does.[5]

- **Listening to calls.** On-device call analysis exists,[10] but reading a customer's call from a payments app is a privacy line I won't cross. I use two call facts only: is a call active, and is the caller a saved contact.
- **Final declines on grey cases.** The model can hold a payment; a person or the customer decides.

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6 The product: Pause

Pause sits inside the payment flow of a UPI app or a bank's Zelle flow. It scores every P2P payment, picks the lightest intervention that will work, and learns from what happens next. My design target is that 99%+ of payments never see anything.

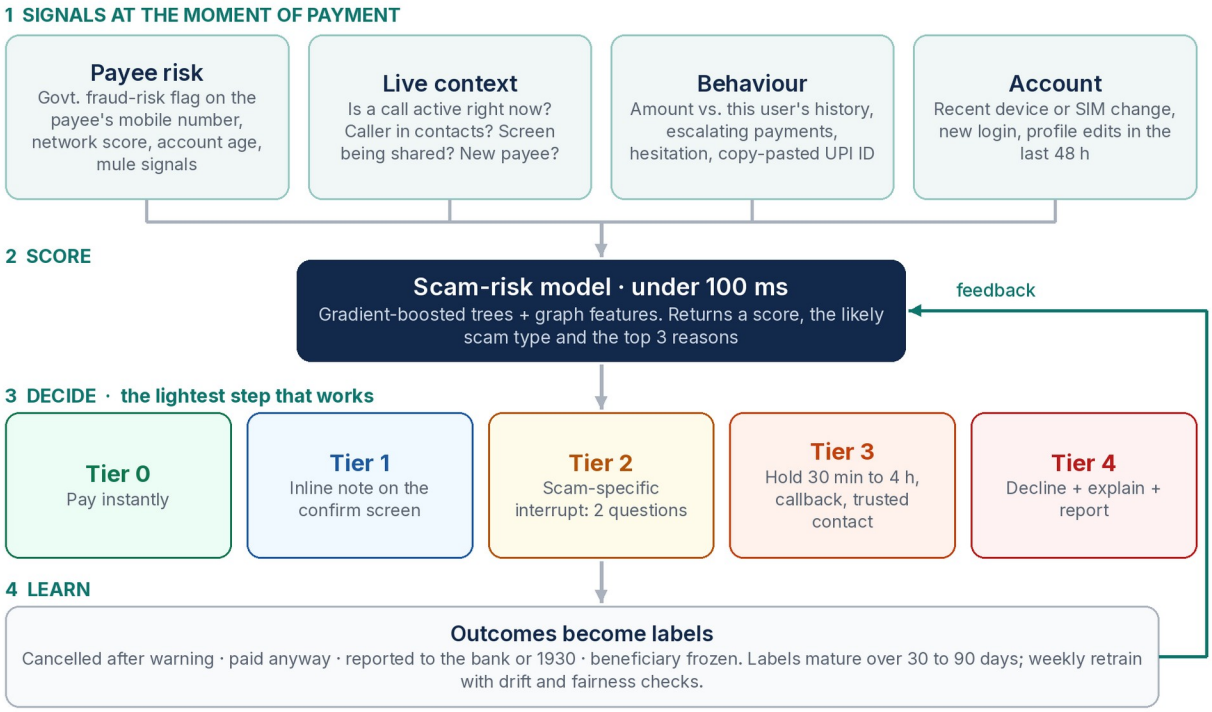


Figure 1. How Pause decides, from tap to outcome. Hard rules sit in front of the model; salary, merchant and mandate payments are never held.

The moment that matters: the Tier 2 interrupt

Most of the product's value is decided on one screen. Here's what Ramesh would see, in Telugu or Hindi (English shown):

> **Wait. Is someone on a call telling you to send this?**
>
> You're on a call with a number that isn't in your contacts, and this is your first payment to this UPI ID. Police, CBI and RBI never ask you to move money to "verify" it.
>
> **1.** Has anyone asked you to keep this payment secret? [Yes] [No]
>
> **2.** Do you know this person outside this call? [Yes] [No]
>
> Scammers often tell people to ignore this screen. If someone told you that, it's a scam.
>
> [Cancel payment] primary [I know them, continue] unlocks after 10 s

- **Questions, not walls of text.** A "yes" on secrecy escalates to a Tier 3 hold. Answering a question about the script is often what breaks it.
- **Specific, not generic.** The first line comes from the model's top reasons. Copy rotates so scripts can't predict the screen.
- **Tier 3 is a hold, not a wall.** A 30-minute to 4-hour countdown, cancel any time, a bank callback on the registered number, and an optional trusted contact. In the US, add an in-app "Is my bank really calling?" check, like Monzo's Call Status.[19]

7 The AI approach

Model choice, build vs buy

Component	Call	Why
Payee risk: DoT's FRI, RBI's DPIP, MuleHunter.AI; Zelle network signals	Consume	No app sees as many payees as the network. DPIP's next phase returns a real-time risk score per transaction.[9]
Behavioural signals: hesitation, coached typing, remote access	Buy (SDK)	Hard to build well; vendors like BioCatch already sell scam-specific signals.[20]
Scam-risk model and policy engine	Build	The loss-vs-conversion trade-off lives here, tuned to each partner.
Intervention UX and copy	Build	This is the product.
Copy drafting and translation	LLM, offline	12+ languages fast, human-reviewed, no live generation risk.

I'd start with gradient-boosted trees: fast enough for the payment path, strong on tabular data, and explainable with reason codes that both the warning and the analyst need. Payee graph features (first-time senders today, links to known mules) are computed off the payment path every few minutes. A sequence model for multi-payment scams like Sneha's comes once there's enough labelled data.

Data and labels

The hardest data problem is the label. **Positives:** payments later reported to the bank or the 1930 helpline, confirmed by a beneficiary freeze where possible. **Weak positives:** payments cancelled after a warning, biased toward what the old model flagged. **Negatives:** no report after 90 days, so the model always learns from last quarter's scams. For blind spots, a small random sample of low-risk payments gets a light Tier 1 note.

Human in the loop. Tier 3 holds go to the partner's fraud team with the score, scam type and top reasons. The analyst calls the customer, and that outcome is our best label, so logging it is one tap. Team capacity caps holds per hour; when the queue is full, the next risky payment gets Tier 2 instead.

Evals and the quality bar

Offline, I judge the model on value-weighted recall at a fixed good-user interrupt rate, not AUC. The bank's real question: if we interrupt one honest payment in 200, how much scam money do we catch? My launch bar (my targets, to validate in shadow mode):

- Good-user interrupt rate (Tier 2+) at or below 0.5% of P2P payments to new payees

- Value-weighted recall of 50%+ on confirmed scams; added p99 latency under 100 ms
- No age band, language or state with an interrupt rate above 2x average without a documented reason

Online: four to six weeks of shadow mode (score everything, show nothing), then a staged rollout. I won't run an A/B test that knowingly shows nothing on a high-risk payment. Experiments live in the middle band: which tier, which copy, how long a hold.

Failure modes and guardrails

Failure mode	Guardrail
False positives on big honest payments (rent, hospital, money home)	Never hold salary, merchant or mandate payments. Trust a payee after one verified payment. Cap interrupts per user per month.
Coached victims ("just press continue")	Copy names the tactic; a "yes" on secrecy escalates to a hold.
Scammers adapt (splitting amounts, switching rails)	Features across the user's recent payments on all rails; weekly retrain; monthly red-team of new scripts.
Signal outage	Fail open for Tiers 0–1, cache hard-rule lists, alert on feed uptime.
Unfair friction on older users	Age is never a model input; outcomes monitored by age band; trusted contact is opt-in.

Responsible AI and regulation

India. Device and call signals need clear consent and purpose limits under the Digital Personal Data Protection Act, 2023. The RBI's 2025 authentication framework already lets issuers add risk-based checks beyond two factors.[1] From 1 January 2027, revised RBI rules count a payment approved "under coercion or duress" as fraudulent, with compensation of 85% of the loss up to ₹25,000 (losses up to ₹50,000, once per lifetime). The RBI funds most of it, but the customer's bank and the beneficiary's bank each pay a share.[7] For the first time a bank has direct money on the line for a digital arrest victim.

US. Regulation E covers unauthorized transfers, not ones people were tricked into sending. Zelle's network rules require reimbursement for certain qualifying imposter scams.[12] The CFPB dropped its case, but New York's is moving.[15, 16] The flip side: a misleading warning or a block with no appeal is its own consumer-protection risk, so every intervention says why and every hold can be cancelled.

8 India vs US: what changes

	India	United States
Who owns the screen	Third-party apps: PhonePe, Google Pay and Paytm carry 90%+ of UPI volume[5]	The customer's bank; Zelle is inside ~2,400 banks and credit unions[12]
Shared signals	FRI on phone numbers[5], MuleHunter.AI in 26 banks[8], DPIP live with 8 banks[9]	Signals inside the Zelle network; I found no public national phone-number risk feed
Economics	P2P UPI is free to users, so friction has no fee to offset; average payment ~₹1,220[2]	RTP averages $3,750 a payment[17]; trust is worth more per deposit customer

India, what I'd do. Go B2B2C through a big app or a large bank. Plug in FRI and DPIP first; they're free and the regulator wants them used. Voice-first warnings in regional languages. Trusted contact as opt-in for older users, not an age mandate.

US, what I'd do. Sell to mid-size banks and credit unions; the biggest banks build in-house. Start with the "move your money to a safe account" script, add the "Is my bank really calling?" check, and frame ROI around reimbursement costs and litigation risk.

9 Success metrics

The north star is scam value stopped, not "scams flagged": flags are cheap, and money is what customers lose. The guardrails stop the team winning by interrupting everyone.

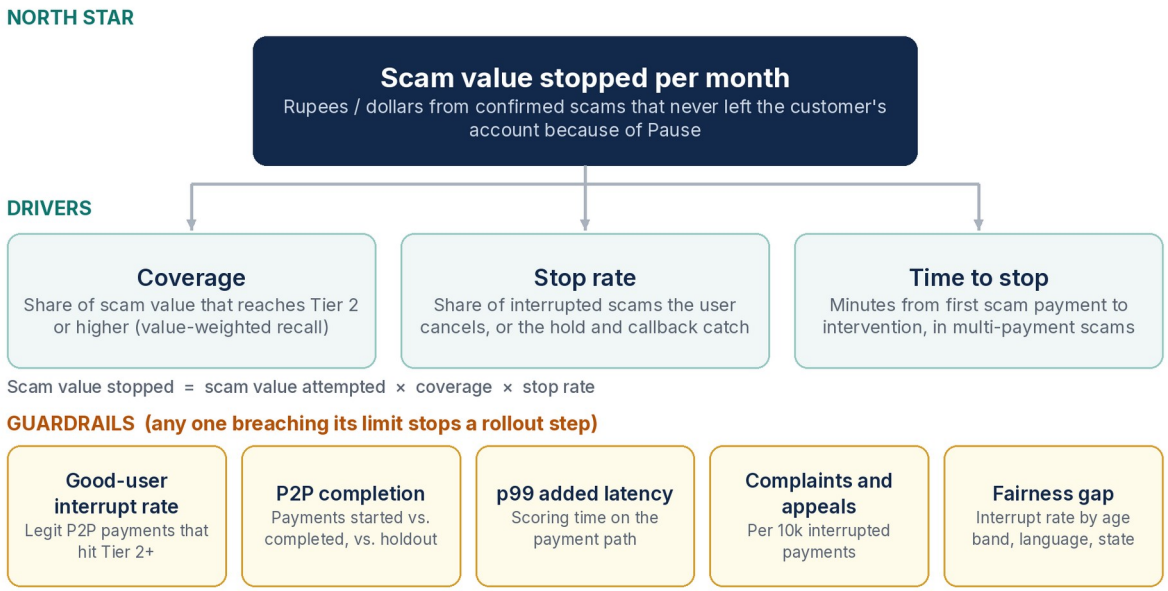


Figure 2. Metrics tree: one north star, three drivers, five guardrails that gate every rollout step.

10 MVP scope and roadmap

Phase	Scope	Exit criteria
MVP **0–3 months**	One partner, Android only (about 96% of Indian phones[10]). P2P to new payees above ₹2,000. Hard rules + FRI + model v0. Tiers 0–2. Hindi, Telugu, English. 4–6 weeks of shadow mode.	Launch bar met in shadow mode; partner signs off on interrupt rate
Phase 2 **3–6 months**	Tier 3 holds with analyst callback (cut from the MVP: I want proof Tier 2 works before a partner staffs a team), trusted contact, DPIP scores, more languages, voice warnings.	Stop rate up, complaints flat, callback SLA met
Phase 3 **6–12 months**	Beneficiary-side scoring for mule accounts, iOS, sequences across UPI and IMPS, US pilot with a credit union group on RTP or FedNow.	Value-weighted recall 60%+; US shadow results

11 Go-to-market

India. First design partner: a mid-size bank or one of the top three apps. The pitch: from 2027 both banks pay into every compensation claim;[7] the RBI is choosing between a blanket lag and risk-based friction, and live numbers from a risk-based system are the strongest input to that choice; and trust is the whole UPI business. Pricing: a fee per 1,000 scored payments plus a share of verified savings (my proposal).

US. Credit unions and community banks. FedNow has 1,725 participants, 19.7% of US institutions,[17] and in my read most lack a scam-modelling team. Sell through their digital banking providers, with a 30-day shadow-mode report on their own traffic as proof.

12 Risks and how I'd de-risk them

Risk	How I'd de-risk it
Partner won't accept any conversion hit	Shadow mode on their data first, weekly completion vs. holdout, interrupt-rate cap in the contract.
The RBI mandates a blanket lag anyway	Pause still helps inside the lag: who to call back, what to say, which payees are safe to whitelist.
Labels too sparse or too late	Pool FRI and DPIP signals, treat analyst outcomes as fast labels, start with the most-reported scam types.
Privacy backlash	On-device yes/no facts only, a clear consent screen, no audio, a public page listing what we read.

13 What I'd measure in the first 90 days

- **Days 1–30, baseline.** Confirmed scams per month at the partner, label delay, share with a beneficiary freeze; baseline P2P completion and loss rate. Ship the hard rules on day one.
- **Days 31–60, shadow mode.** Value-weighted recall at 0.25%, 0.5% and 1% interrupt rates; precision by scam type; p99 latency on live traffic.
- **Days 61–90, first live users.** Tiers 1–2 for 10% of eligible users. Stop rate by copy variant, completion vs. holdout, complaints per 10,000 interrupts, interrupt rate by age, language and state. Day 90 ends in one decision: scale, retune or stop.

> **The trade-off I'd defend in an interview**
>
> I'd rather ship a model that interrupts one honest payment in 200 and explains itself than one that catches a little more and teaches people that instant payments can't be trusted. The goal isn't more friction. It's friction in the right place.

Sources

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Estimates and unverified figures. My estimates: ~840,000 Zelle reports a year (0.02% × 4.2 bn), the ₹115 crore year-one figure, the launch-bar targets and the 99%+ Tier 0 target. Unverified on the source page: the ~$2 bn FTC bank-transfer figure and the 756-bank count. Digital arrest figures come from secondary reports.